Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Val

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Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values

by

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1 Introduction and Background

Health insurance cost prediction is an important problem in actuarial sciences, health care analytics and applied statistical modelling. Insurers need to estimate expected medical costs or premiums using information about individuals, including demographic, health-related, lifestyle, occupational, regional and policy-related characteristics. These characteristics may influence insurance costs in complex and nonlinear ways, especially when variables interact with one another.

From a statistical learning perspective, the proposed study is a supervised learning problem. The dataset contains observed responses together with explanatory variables, and the aim is to learn a relationship that can be used to predict future responses. In this study, the observed response is the health insurance charge or premium, while the explanatory variables are policyholder, health, lifestyle, socioeconomic and insurance-plan characteristics. Since the response variable is continuous, the task is a regression problem rather than a classification problem. This distinction is important because the model must be evaluated using regression metrics such as root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination, R^2, rather than classification accuracy. This framing follows the statistical learning view that supervised learning seeks to predict a response from a set of explanatory variables, with regression used for continuous responses and classification used for categorical responses (James et al., 2021; Hastie et al., 2009). The use of RMSE, MAE and R^2 is also consistent with recent medical insurance cost prediction studies that evaluate continuous-cost prediction models using regression-based performance measures Orji and Ukwandu (2024).

Traditional statistical models, such as regression-based approaches, have been widely used for insurance pricing because they are interpretable and relatively easy to validate. However, modern insurance datasets may contain nonlinear relationships and interactions that are not easily represented by simple linear structures. Artificial neural networks provide a flexible class of nonlinear statistical models. They can be written conceptually as

$$Y_i = f_{NN}(\mathbf{x}_i, \theta) + \epsilon_i,$$

Where $\mathbf{x}_i$ is a vector of predictors, Y_i is the response, θ represents the network parameters and ϵ_i is the error term or random noise term for observation i. This formulation motivates their use in a health insurance setting where cost is likely to depend on multiple interacting risk factors.

Several previous studies have applied machine learning methods to health insurance premium or cost prediction. For example, Kulkarni et al. (2022) developed a machine learning-based regression framework to predict health insurance premiums framework to using variables such as age, gender, BMI, children, smoking habits, region, medical history, family medical history, exercise, employment status and coverage level. Their study used an artificial neural network and evaluated performance using regression metrics such as RMSE, MSE, MAE, R^2, and adjusted R^2. Other studies, including those by Bhatia et al. (2022), Narayana et al. (2023), Chittilappilly et al. (2023), Islam et al. (2023), and Kulkarni et al. (2022), have also explored medical insurancw cost prediction, health insurance premium prediction, or insurance claim prediction using machine learning methods.

Although these studies show that machine learning can be useful in health insurance cost prediction, many of them focus mainly on predictive accuracy. Less attention is given to understanding how the fitted model behaves under many possible input scenarios, and to explaining the drivers of the model's predictions. This is important because a deep neural network may produce accurate predictions, but the model may still be difficult to interpret.

The deep neural network will be used to model health insurance cost predictions. Monte Carlo simulation will be used after the final DNN has been selected to generate synthetic input profiles and analyse the distribution of predicted costs. Shapley values will then be used to interpret the contribution of each input variable to the final DNN predictions. The purpose of the study is therefore not only to train a DNN, but to understand the distribution and drivers of the model's health insurance cost predictions.

2 Research Landscape

Machine learning has become increasingly relevant in health insurance cost prediction. In this setting, the response variable is usually a continuous insurance charge, premium, or medical cost. This makes the problem a regression problem rater than a classification problem. Kaushik et al. (2022) provide a useful example of machine learning applied to health insurance premium prediction. Their study used a dataset with about 1M entries and twelve columns, including age, gender, BMI, children, smoking habits, region, medical history, family medical history, exercise, employment status, coverage level and charges. The authors trained an artificial neural network model and evaluated it using standard regression metrics.

The proposed study uses a larger health insurance dataset. The selected Kaggle dataset, *Insurance Data for Machine Learning*, is described as a US health insurance premium prediction dataset containing one million generated records. The dataset is listed under a CC0 public-domain licence Sridhar Streaks (2023).

Deep neural networks are flexible models that can learn nonlinear relationships from data. However, tabular data remains a challenging domain for deep learning, and the use of DNNs on structured tabular datasets must be justified through careful empirical evaluation rather than assumed to be automatically superior Borisov et al. (2022). Goodfellow et al. (2016) also emphasise that a neural network should be treated as one candidate model class among others: the structure of the network, its objective function and its out-of-sample performance determine whether it is a useful representation of the data.

In addition to prediction, this study incorporates interpretability through Shapley values. Lundberg and Lee (2017) proposed SHAP as a unified framework for interpreting predictions from complex models by assigning each feature an importance value for a particular prediction. This is relevant to the proposed thesis because DNN predictions may be difficult to explain without post-model interpretation.

Existing studies have applied machine learning and neural network methods to health insurance costs or premium prediction, but there remains scope to combine DNN-based prediction with Monte Carlo input simulation and Shapley-value interpretation on a larger US health insurance dataset. This combination can help explain not only what the model predicts, but also how the predicted costs are distributed and which variables drive those predictions.

3 Problem Statement

The main problem addressed in this research is the gap between *prediction* and *understanding* in health insurance cost modelling. Existing health insurance cost prediction studies often report point-prediction performance using metrics such as RMSE, MAE, and R^2. However, point prediction alone does not explain how predicted costs behave under many possible input profiles. It also does not explain which variables are responsible for high or low predicted costs.

This study therefore addresses three connected issues:

1. *Prediction*: How can a DNN be used to predict health insurance costs from the available input variables?

2. *Simulation*: How can Monte Carlo simulation be used to generate synthetic input profiles and study the distribution of DNN-predicted insurance costs?

3. *Interpretation*: How can Shapley values be used to explain the contribution of each input variable to the final DNN predictions?

The proposed framework aims to move beyond a simple prediction exercise by combining model fitting, simulation-based analysis and explainable machine learning.

4 Research Questions

The primary research question guiding this study is stated as follows:

> *How can deep neural networks, Monte Carlo simulation and Shapley values be combined to predict and interpret health insurance costs?*

To address the primary research question, the following supporting research questions are considered:

1. What pre-processing steps are required to prepare the health insurance dataset for deep neural network modelling?
2. How can a deep neural network be trained to predict health insurance costs from the available input variables?
3. How can Monte Carlo simulation be used to generate synthetic input profiles and study the distribution of predicted costs?
4. How can Shapley values be applied to interpret the contribution of each input variable to the deep neural network predictions?
5. What do the Monte Carlo simulation and Shapley value results reveal about the key drivers of high-cost health insurance predictions?

5 Aim and Objectives

The aim of this study is to develop an intepret a DNN-based framework for health insurance cost prediction using Monte Carlo input simulation and Shapley values.

The following are the objectives:

1. To review previous studies on health insurance cost and premium prediction using machine learning and neural networks.
2. To describe and analyse the selected US health insurance dataset.
3. To investigate the distributional properties of both the input variables and the response variable, including skewness, heavy-tailed behaviour, outliers, variable ranges and categorical imbalance.

4. To pre-process the dataset through suitable encoding, scaling and transformation methods where appropriate.

5. To train and tune a deep neural network regression model for health insurance cost prediction.

6. To evaluate the final DNN model using suitable regression metrics, including RMSE, MAE, R^2, and actual-versus-predicted plots.

7. To use Monte Carlo simulation to generate synthetic input profiles after the final DNN has been selected.

8. To analyse the distribution of predicted health insurance costs produced by the trained DNN under simulated input profiles.

9. To apply Shapley values to interpret the contribution of each input variable to the DNN predictions.

6 Data Description

The study will use the publicly available *Insurance Data for Machine Learning* dataset from Kaggle. The dataset is described as a US health insurance premium prediction dataset containing one million generated records. It includes variables related to demographic, health, lifestyle, socioeconomic and insurance-plan characteristics.

The response variable is the health insurance charge or premium:

$$Y = \text{ health insurance charge or premium.}$$

The explanatory variables include factors such as those shown in Table 1.

Variable category	**Examples**
Demographic	Age, gender, region
Health-related	BMI, medical history, family medical history
Lifestyle	Smoking status, exercise frequency
Occupational	Occupation
Policy-related	Number of children/dependants, coverage level

Table 1: Summary of variable categories in the proposed dataset.

After categorical variables are encoded into numerical form, the model-ready dataset is expected to contain approximately 22 input columns and one response variable. The study will examine both the response variable and all input variables. The exploratory analysis will include missing values, outliers, skewness, heavy-tailed behaviour, categorical frequencies, variable ranges and scaling requirements. All available explanatory variables will initially be treated as candidate predictors. The study will not assume in advance that any variable is irrelevant.

7 Proposed Methodology

The proposed methodology consists of four main stages:

1. exploratory data analysis and preprocessing;
2. DNN model training and selection;
3. Monte Carlo input simulation; and
4. Shapley-value interpretation.

7.1 Exploratory Data Analysis and Preprocessing

The study will begin with exploratory data analysis of the full dataset. This will include descriptive statistics and graphical summaries for both the input variables and the response variable. The analysis will investigate missing values, outliers, skewness, heavy-tailed behaviour, categorical distributions and variable ranges.

Categorical variables will be encoded into numerical form. Numerical variables will be scaled or normalised where appropriate after examining their distributional properties. If the response variable is highly skewed, a transformation such as a logarithmic transformation may be considered, with final evaluation still reported on the original monetary scale.

The dataset will be split into training, validation and test sets. The training set will be used to fit the DNN model, the validation set will guide model selection, and the test set will be reserved for final evaluation. This split is important because the purpose of model fitting is not merely to reduce training error, but to produce a model that generalises to unseen data. The validation framework follows the standard bias-variance tradeoff discussed by Hastie et al. (2009), where model complexity must be balanced against out-of-sample performance.

7.2 Deep Neural Network Modelling

The DNN will be formulated as a supervised regression model. Let the dataset be:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N},$$

where $\mathbf{x}_i \in \mathbb{R}^{22}$ is the encoded input vector and $y_i \in \mathbb{R}$ is the observed health insurance charge or premium. The DNN learns a function:

$$\widehat{y}_i = f_\theta(\mathbf{x}_i),$$

where θ represents the network weights and biases.

Feed-forward neural networks can be represented as recursive nonlinear transformations. For input

$$\mathbf{h}^{(0)} = \mathbf{x},$$

the hidden layers are written as

$$\mathbf{h}^{(\ell)} = \sigma\big(W^{(\ell)}\mathbf{h}^{(\ell-1)}\big), \qquad \ell = 1, \ldots, L,$$

and the final regression output is

$$\widehat{y} = W^{(L+1)}\mathbf{h}^{(L)}.$$

The activation function $\sigma(\cdot)$ introduces nonlinearity in the hidden layers. Common activation functions include logistic, hyperbolic tangent and rectified linear unit functions. In this study, ReLU-type hidden activations are suitable because they are widely used in modern DNNs and allow efficient learning of nonlinear relationships. Since the response variable is continuous, the output layer will use a linear activation. This is appropriate for regression, whereas classification would typically use logistic or softmax output functions.

An initial feed-forward architecture will be considered, beginning with 22 input neurons and one output neuron:

The output layer has one neuron because the response is a continuous insurance charge or premium. The final DNN architecture will be selected using training and validation performance.

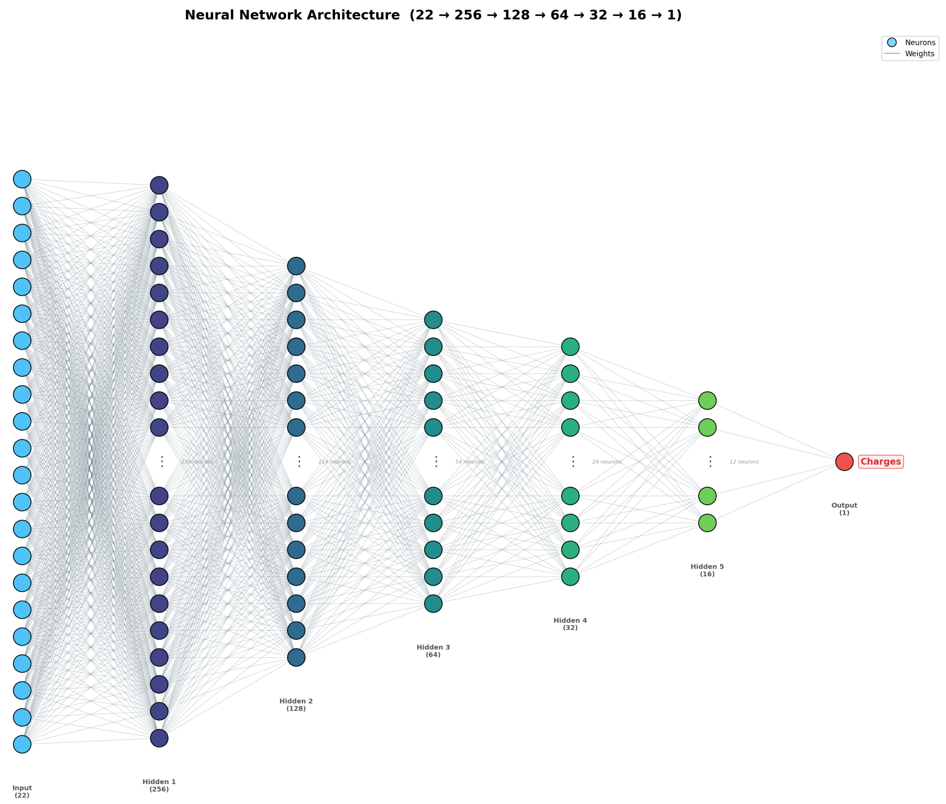

Figure 1: Neural network architecture.

7.3 Training, Optimisation and Backpropagation

The DNN will be trained using mean squared error:

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2.$$

For a continuous response, MSE is an appropriate training objective because it penalises larger prediction errors more strongly and aligns naturally with regression modelling. The network parameters are estimated through gradient-based optimisation. In general, the parameter update can be written as

$$\theta^{(m)} = \theta^{(m-1)} - \eta \nabla_\theta \mathcal{L}\big(\theta^{(m-1)}\big)\,,$$

where η is a learning-rate or step-size parameter and $\mathcal{L}(\theta)$ is the training loss. Backpropagation provides an efficient method for computing the gradients of the loss function with respect to the weights and biases in the network. Detailed discussion of the DNN architecture, activation functions, optimiser and training diagnostics will be provided in the full thesis methodology chapter.

7.4 Model Evaluation

The final selected DNN will be evaluated on the test set using regression metrics:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2},$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \widehat{y}_i|,$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}.$$

An actual-versus-predicted plot will also be used to visually assess model performance. For each test observation, the predicted cost $\widehat{y}_i$ is plotted against the true cost y_i, with the 45-degree line $\widehat{y} = y$ overlaid. A perfect model places all points on this line.

If the response variable is transformed during training, final predictions will be back-transformed and all metrics evaluated on the original monetary scale. Since this is a regression problem, the study will avoid treating performance as classification accuracy. If the term "accuracy" is used informally, it will be interpreted in relation to regression performance, especially R^2.

7.5 Monte Carlo Input Simulation

After the final DNN model has been selected, Monte Carlo simulation will be used to generate synthetic input profiles. Let the input vector be:

$$\mathbf{X} = (X_1, X_2, \dots, X_p).$$

Monte Carlo simulation will generate M input profiles:

$$\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots, \mathbf{X}^{(M)}.$$

Each simulated input profile will be passed through the trained DNN:

$$\widehat{Y}^{(m)} = f_{\widehat{\theta}}\big(\mathbf{X}^{(m)}\big), \qquad m = 1, \dots, M.$$

This produces a simulated distribution of predicted health insurance costs:

$$\widehat{Y}^{(1)}, \widehat{Y}^{(2)}, \dots, \widehat{Y}^{(M)}.$$

The simulated predicted costs will be analysed using mean, median, standard deviation, interquartile range, upper quantiles, histograms and density plots. The Monte Carlo component will therefore allow the study to examine the model-implied distribution of health insurance costs, rather than only individual point predictions.

7.6 Shapley-Value Interpretation

Shapley values will be used to interpret the final DNN predictions. In machine learning, Shapley values assign each input variable a contribution to a model prediction. For a prediction $f(\mathbf{x})$, the SHAP decomposition can be written as:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{p} \phi_j,$$

where ϕ_0 is the baseline prediction and ϕ_j is the contribution of input variable j.

The study will use Shapley values in two ways. First, global interpretation will identify which variables contribute most strongly to predictions across the dataset. Second, local interpretation will explain individual predictions, especially high-cost predictions identified through Monte Carlo simulation. Shapley values will be interpreted as explanations of the fitted model, not as causal effects.

8 Expected Contributions

This study is expected to make the following contributions:

1. It will develop a DNN-based framework for health insurance cost prediction using a larger US health insurance dataset.
2. It will provide a distributional analysis of DNN-predicted health insurance costs through Monte Carlo input simulation.
3. It will use Shapley values to interpret the contribution of input variables to the final DNN predictions.
4. It will connect prediction, simulation and interpretation into one coherent framework.
5. It will provide a statistically grounded approach for understanding model-implied health insurance cost predictions, rather than relying only on point predictions.

The study does not claim that DNNs are always superior to other machine learning models. Instead, it investigates how a DNN can be trained, simulated and interpreted for the selected health insurance cost prediction problem.

9 Limitations

The study will use a publicly available dataset, which may not fully represent the complexity of real-world insurance pricing. Real underwriting systems may include additional variables such as claims history, contractual details, exclusions, underwriting rules and regulatory constraints.

The Monte Carlo simulation will generate synthetic input profiles based on the available data and sampling logic. These simulated profiles are useful for scenario analysis, but they do not create new real-world observations. Shapley values explain the fitted model's predictions, but they do not establish causality. If a feature has a high Shapley contribution, this means the model relies strongly on that feature in prediction; it does not prove that the feature causally determines insurance cost.

10 Ethical Considerations

The study will use a publicly available dataset and will not involve direct human participants. The dataset is listed as publicly available under a CC0 public-domain licence.

Although the dataset is public, the study will still consider fairness and interpretability issues. Health insurance pricing may involve sensitive or potentially sensitive variables, such as age, gender, region, health-related indicators and occupation. The use of Shapley values will assist in understanding how strongly these variables influence predictions.

The study will avoid presenting the model as a final pricing system for real-world deployment. Instead, it will be treated as an academic modelling framework for understanding health insurance cost predictions.

11 Timeline and Work Plan

Period	Activity
May 2026	Finalise research proposal, dataset source and initial reference list.
June 2026	Conduct detailed literature review and refine Chapter 2 outline.
July 2026	Perform exploratory data analysis and investigate distributions of all variables.
August 2026	Complete preprocessing, encoding, scaling decisions and data splitting.
September 2026	Train initial DNN models and begin model tuning.
October 2026	Select final DNN model and evaluate on test data.
November 2026	Conduct Monte Carlo input simulation and analyse predicted cost distributions.
December 2026	Apply Shapley-value interpretation to final DNN predictions.
January 2027	Write results and discussion chapters.
February 2027	Complete full thesis draft.
March 2027	Revise thesis based on supervisor feedback.
April 2027	Final editing, formatting and submission preparation.

Table 2: Proposed timeline and work plan.

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